

Crime Rate Prediction for the City of Los Angeles from 2020-2025

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1 Introduction

The dataset consists of crimes reported and arrest data from the City of Los Angeles from 2020 to 2025. Using reported crime data from the Los Angeles Police Department (LAPD), this study aims to apply a statistical model to predict whether a crime is violent or non-violent based on location, victim characteristic, and time of occurrence.

2 Objectives

The objective of this study is to develop a statistical model to predict whether a reported crime is violent or non-violent based on the victim, location and time of occurrence.

3 Purpose

The purpose of this model is to predict whether a reported crime is violent or non-violent and, therefore, help the Officials, the city, and law enforcement improve their safety strategies and prevent or minimize crime rate

4 Dataset Details

The proposed data set is from the (LAPD) and includes crime incidents reported from 2020-2025. It provides details on crime occurrence time and date, location information, and victim demographics such as age and sex. The model utilizes only those related to the victim, location, and timing of the incident. The response variable is the part 1 and part 2 crime classification, which distinguishes violent crimes from non-violent crimes and is used to assess model performance.

Number of records:1004992 Number of parameters:28

Source: <https://catalog.data.gov/dataset/crime-data-from-2020-to-present>

No.	Variable Name	Role	Type	Description	Units
1	DR_NO	Identifier	Text (Identifier)	Unique division of record numbers assigned to each crime.	-
2	Date Rptd	Predictor	Date / Timestamp	Date when the crime was reported.	Date
3	DATE OCC	Predictor	Date / Timestamp	Date when the crime occurred.	Date
4	TIME OCC	Predictor	Time	Time the crime occurred (24-hour format).	Time
5	AREA	Predictor	Text	Code representing one of the 21 LAPD community police stations.	-
6	AREA NAME	Predictor	Text	Name of the LAPD area referencing the surrounding community.	-
7	Rpt Dist No	Predictor	Text	Code representing a sub-area with geographic significance used for statistical comparison.	-
8	Part 1-2	Target/Response	Categorical	Classification of crimes as Part 1 (violent) or Part 2 (non-violent).	-
9	Crm Cd	Predictor	Number	Code identifying the crime committed.	-
10	Crm Cd Desc	Predictor	Text	Description corresponding to the crime code provided.	-
11	Mocodes	Predictor	Text	Describes activities and methods used to carry out the crime.	-
12	Vict Age	Predictor	Numeric	Age of the victim at the time of the crime.	Years
13	Vict Sex	Predictor	Categorical (Nominal)	Sex of the victim (F = Female, M = Male, X = Unknown).	-
14	Vict Descent	Predictor	Categorical (Nominal)	Code representing the victim's racial or ethnic descent.	-
15	Premis Cd	Predictor	Categorical (Nominal)	Numeric code identifying the type of premises or location.	-
16	Premis Desc	Predictor	Text	Description of the premises where the crime occurred.	-
17	Weapon Used Cd	Predictor	Categorical (Nominal)	Code identifying the weapon used in the crime, if any.	-
18	Weapon Desc	Predictor	Text	Description of the weapon used during the crime.	-
19	Status	Predictor	Categorical (Nominal)	Code indicating the status of the crime investigation.	-
20	Status Desc	Predictor	Categorical (Nominal)	Description of the investigation status.	-
21	Crm Cd 1	Predictor	Categorical (Nominal)	Code for the primary and most serious crime committed.	-
22	Crm Cd 2	Predictor	Categorical (Nominal)	Code for an additional, less serious crime if applicable.	-

23	Crm Cd 3	Predictor	Categorical (Nominal)	Code for an additional, less serious crime if applicable.	-
24	Crm Cd 4	Predictor	Categorical (Nominal)	Code for an additional, less serious crime if applicable.	-
25	Location	Predictor	Text	Street address of the crime incident.	-
26	Cross Street	Predictor	Text	Nearest cross street to the crime location.	-
27	LAT	Predictor	Number	Latitude coordinate of the crime location.	Degrees
28	LON	Predictor	Number	Longitude coordinate of the crime location.	Degrees

Description of variables in the LAPD crime dataset (2020–2025)